

COMPSCI4087

Startup Growth Engineering

Comprehensive Exam Study Notes

University of Glasgow | 2025-26

Closed-book exam: 105 minutes | 60 marks | 16 questions

Topic	Marks	% of Exam	Priority
Activation & Retention	22	37%	HIGHEST
Compounding Growth	17	28%	HIGH
Acquisition	15	25%	MEDIUM
Organisation	4	7%	LOW
Investment	2	3%	LOW

Products confirmed for the exam: ChatGPT, Snapchat, Medium, Food Delivery (Deliveroo/Uber Eats/Meituan)

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Chapter 1: Foundations of Growth Engineering

1.1 What is Growth Engineering?

Growth Engineering is the systematic, data-driven approach to growing a start-up to scale. It asserts that start-up growth is not purely luck but is largely governed by repeatable techniques tested and evolved in tech ecosystems like Silicon Valley.

A start-up's growth follows one of three trajectories: **exponential growth** (the ideal), **linear growth** (sustainable but limited), or **failure** (losing users faster than gaining them). Growth Engineering provides the tools and frameworks to aim for the first.

1.2 The Four Pre-Cursors to Growth Engineering

Four 'journeys' in industrial product engineering created the conditions for Growth Engineering to exist:

1. Delusion of Certainty --> Embracing Uncertainty

New product introduction is inherently uncertain. The **Lean Startup methodology** evolved to help teams work amidst this uncertainty:

- **Fail-Fast:** Since ~80% of startups fail, discover failure as quickly and cheaply as possible.
- **Pivot-or-Persevere:** Regular discipline of asking: is the product making progress toward adoption? If not, change direction (pivot) or continue (persevere).
- **Learning Metrics vs Vanity Metrics:** Learning metrics reveal whether users are genuinely engaging (e.g. Monthly Active Users). Vanity metrics look impressive but mask real performance (e.g. total app downloads).
- **Product-Market Fit (PMF):** The state where the market wants your product. Do NOT scale marketing before achieving PMF — you will waste money acquiring users you cannot retain.

KEY CONCEPT: A common fatal mistake: assuming PMF and prematurely scaling marketing. It is always better to retain a user than to lose that user and have to acquire a new one.

2. Functional Pre-Eminence --> Model Pre-Eminence

Traditional companies organise by function (Marketing, Engineering, Product). Growth Engineering requires teams to work seamlessly across these boundaries because growing a product requires multiple skillsets working together on the customer journey.

KEY CONCEPT: Conway's Law: 'You ship your organisation.' An organisation's communication structure is reflected in the products it builds. Internal team boundaries become boundaries in the product. If Marketing, Product, and Engineering operate in silos, the customer journey will have seams between those functions.

3. Low-Speed Iteration --> High-Speed Iteration

Tech companies now release weekly or daily, versus once or twice a year 20 years ago. Faster iteration means faster experimentation, faster learning, and faster progress toward PMF.

KEY CONCEPT: Boyd's Law of Iteration: Speed of iteration beats quality of iteration. The entity that iterates faster will learn faster and outperform competitors. BUT — iterating without a directing strategy is wasteful. 'If you are digging your grave, you do not want to dig faster — you want to stop digging.'

4. Metric Soup --> Model-Based Metrics

Problems with metrics in businesses: (1) an explosion of metrics that obscure insight ('metric soup'), and (2) making the **measurable** important rather than making **what is important** measurable. Growth Engineering addresses this by building comprehensive models of how the business works and measuring what matters within that model.

EXAM TIP: Know the distinction between Learning Metrics and Vanity Metrics. Learning metrics express a measure of **activity** and are therefore **frequency-based**. Vanity metrics are simple accumulators with no relationship to frequency (e.g. total downloads).

Chapter 2: The Growth Model

2.1 Strategy = 'How Do We Grow?'

In start-up terms, strategy is the answer to the question: **How do we grow?** If you ask each employee of a start-up this question and receive multiple different answers (or no clear answer), the start-up has no clear strategy. The Growth Model IS the practical implementation of that strategy.

2.2 The Three-Phase High-Level Strategy

Every start-up's highest-level strategy involves three sequential phases:

- **Phase 1 — Find Product-Market Fit:** Determine that the market genuinely wants and will adopt your product.
- **Phase 2 — Exploit PMF:** Grow the market opened by having PMF, as aggressively as possible.
- **Phase 3 — Move to Adjacencies:** As the market saturates, identify new geographies, use cases, or user domains and repeat.

2.3 The Three Components of the Growth Model

The Growth Model has three interconnected components that, together, define how the business grows:

Component 1: Optimise the Customer Journey

Users pass through five states: **Acquire** → **Activate** → **Retain** → **Monetize** (and **Refer** new users). The goal is to move users efficiently from acquisition through to habitual use, monetisation, and referral.

Component 2: Drive Compounding Growth

Mechanisms that cause users to refer other users, creating 'free' acquisition. When the rate of growth is proportional to the current number of users, growth compounds. This is driven by **Compounding Growth Mechanisms (CGMs)** and **Network Effects**.

Component 3: Linear Marketing

Traditional marketing (SEO, SEM, PR, ads) that brings users to the product for the first time. Linear marketing seeds the compounding growth mechanisms.

KEY CONCEPT: Analogy 1: Linear Marketing is the flame that ignites Compounding Growth Mechanisms.

Analogy 2: Linear Marketing is the Principal; Compounding Growth is the Interest.

Without linear marketing, CGMs have no initial users to operate upon. Some rare products (with extremely powerful organic word-of-mouth) can grow without classical linear marketing, but most start-ups need it.

2.4 The Customer Journey in Detail

Acquisition: Making users aware of the product for the first time (visiting the website, downloading the app). These people are NOT yet users.

Activation: Getting the user into the **habit** of using the product for the first time. If users do not form a habit, they are lost. This is the critical bridge between acquisition and retention.

Retention: Keeping users in the habit state. Only retained users can be monetised and can refer others. Retention is arguably the most important state of all.

Monetization: Generating revenue — subscriptions, purchases, advertising, etc.

Referral: Encouraging users to tell others about the product. This creates a new, essentially free acquisition channel and is the foundation for compounding growth.

EXAM TIP: Retention is the most important state because it enables both monetisation and referral. Poor retention will dominate eventually, no matter how good your marketing is. Be ready to explain WHY retention matters more than acquisition.

Chapter 3: Retention — The Most Important Metric

3.1 Why Retention Dominates Growth

Retention's influence on overall growth is far more profound than it appears at first glance. Consider the 'Jumping the Shark' pattern (e.g. FarmVille) — rapid rise followed by collapse. This shape is dominated by retention failure.

Key insights from retention modelling:

- Even **1% churn per period** causes the active user count to peak earlier and then decline.
- **5% churn** has a very severe impact and creates a shark-fin growth curve.
- No matter how sophisticated marketing is, and no matter how viral a product is, **retention wins in the end** — poor retention will dominate eventually.
- Retention also directly impacts **referral** (and therefore compounding growth) and **revenue**.

Comparison: with 0% churn and 10% referral, a product reaches ~300k users. With 20% churn and the same referral rate, it reaches only ~4 million (counterintuitive until you consider the compounding lost from churned users not referring).

Wait — actually, 0% churn reaches 4 million and 20% churn reaches only ~300k. The better retention gives the vastly larger outcome because retained users keep compounding referrals.

3.2 The Retention-Cohort Graph

This is **probably the most important chart in the start-up world**. It shows, for each cohort of users, the percentage still active over time. Three typical patterns:

- **Green line (Viable business):** Drops during activation then **levels off** — a 'retention floor.' Users have formed habits. If most cohorts look like this, you have achieved PMF.
- **Yellow line (Work to do):** Continues to fall slowly after activation. Not catastrophic but needs improvement. Very common for early-stage startups.
- **Red line (Heading for failure):** Continues to fall rapidly. Unless significantly improved, the startup will not survive.

KEY CONCEPT: If a start-up regularly plots cohort retention charts and uses them to guide product and marketing activities, it has a far better chance of finding PMF. The retention-cohort graph is the most important chart in the start-up world.

3.3 The Retention Metric — NaP

To draw retention charts, you need to define what an 'active user' means. The retention metric (also called habit metric or active user definition) has the form:

NaP = N actions of type 'a' performed in period P (days)

Examples:

- Skyscanner: **2b365** (2 bookings every 365 days)

- Deliveroo: **4o28** (4 orders every 28 days)
- Uber: **2u7** (2 rides every 7 days)
- Facebook: **1l1** (1 login every day)

Rules for Setting NaP

- **N should be > 1**: Better signal-to-noise ratio. Fewer false churn signals.
- **P should be > the typical usage cycle**: E.g. if the product is used weekdays only, P=1 day would create weekend 'false churn' noise.
- **If P is very large** (e.g. yearly for travel), use a **proxy metric** — less accurate but more responsive. E.g. Skyscanner could use searches instead of bookings.
- **Early-stage**: Not enough data for correlation analysis. Pick best guess and refine over time.

How Growth Teams Choose the NaP Metric

Track every user action at every frequency. Correlate those combinations with whether users are retained 12 months later. The combination that correlates most strongly with future retention becomes the NaP metric. In early stages with insufficient data, use a reasonable estimate and refine.

User Categories Based on NaP

Example for a delivery product (base metric: 4o28):

- **Power user**: (6+)o28 — most valuable, deepest habit
- **Core user**: (4-5)o28 — healthy, retained
- **Churn-risk**: (<4)o28 — trigger to act, intervene to recover

When a user drops from Core to Churn-risk, the team should intervene immediately with targeted actions to recover them.

3.4 Cohort Retention Tables

Three views of the same underlying data, each revealing different insights:

- **A2 — Absolute numbers retained**: Raw customer counts per cohort per month. Good for seeing overall scale.
- **B1 — % retained relative to base month**: Best for comparing cohorts against each other. Normalises out different cohort sizes.
- **B2 — % churned relative to base month**: Clearly identifies which periods have the worst drop-off. Helps pinpoint where in the user lifecycle problems exist.

EXAM TIP: When asked to 'comment on a cohort chart,' first describe the activation drop-off, then describe the retention trend (levelling off? continuing to fall?). Classify as green/yellow/red. Look for anomalies — one cohort out of sequence may indicate a bug, bad experiment, or seasonal effect. If later cohorts improve, attribute to ongoing product improvement.

3.5 Factors Influencing Retention

Retention is increased by increasing **engagement intensity**. The key driver is **natural frequency** — how often users naturally need to use the product:

Frequency	Daily	Weekly	Monthly	Yearly
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Ease of Retention	Easy			Difficult
Example	Social Media	Pinterest	Travel	Tax Software

Two strategies to increase natural frequency:

- **Add new features and use cases** — e.g. Amazon went from books (monthly) to streaming (daily).
- **Increase usage rate of existing features** — e.g. Netflix autoplay, recommendation engines.

For low-frequency products, add **additional awareness frequencies** between buying events (e.g. travel company newsletters). This reminds users the product exists.

Chapter 4: Activation — From Acquisition to Habit

4.1 What is Activation?

Activation is the bridging state between Acquisition and Retention. Its task is to take newly acquired potential users and deliver them into a **habit** around the product. Users who have used the product once or twice are NOT activated. Only when they are in a habitual usage state have they activated.

Activation is responsible for the rapid initial decline in retention-cohort charts. It is natural to lose more users before they have formed a habit than after. But improving activation performance means continuing from a higher retention base.

4.2 The Two Stages of Activation

Stage 1 — Demonstrate Value: Show the user that the product is valuable as quickly as possible. This has two sub-stages: (a) gather minimum required information from the user, and (b) deliver initial value based on that information.

Stage 2 — Build to Habit: Increase frequency of value delivery until the user establishes habitual use at the product's natural frequency.

4.3 The Activation Inequality

$$\text{Perceived Value} > \text{Perceived Effort}$$

If the effort required (signing up, entering details, learning the product) exceeds the perceived value, the user will be lost before activation completes. Growth teams use this concept to design onboarding workflows that front-load value delivery and minimise information gathering.

Only the **minimum information required** to deliver value should be sought from the user. For example, Skyscanner needs to know where you want to fly before offering options — but it does NOT need your credit card details at that point.

4.4 Cognitive Energy Model

Each step in the activation workflow either **depletes** (fatigues) or **increases** (excites) the user's cognitive energy. If cognitive energy reaches zero, the user will not activate. Design the workflow to minimise depleting steps and add energising steps (value demonstrations, 'aha moments').

A common mistake: asking for basic sign-up info, then demonstrating value, then asking for credit card details. If the gap between these requests is too long, it feels like TWO sign-up processes, which increases perceived friction.

4.5 Techniques to Increase Activation Rate

Reminder techniques (to bring users back to the product):

- Push notifications, email prompts, retargeting ads

Friction-reduction techniques (to make finding value easier):

- Recommendation engines, personalised content, simplified onboarding flows

EXAM TIP: If asked 'how do growth teams use the Activation Inequality,' explain: (1) the concept (perceived value > perceived effort), and (2) teams use it to optimise the ordering of steps in the activation process and the amount of information requested before delivering value.

Chapter 5: Acquisition — Channels and Strategy

5.1 Three Types of Acquisition Channels

Intent-Based Channels

Users are actively searching with intent to buy or complete an action. **High conversion rates** because users already want something your product provides.

- **SEO (Search Engine Optimisation):** 'Free' — optimise web pages to rank highly in organic search results. Higher ranking = dramatically better conversion. Below the first page = essentially invisible.
- **SEM (Search Engine Marketing):** Pay-per-click advertising on search engines. Appear as ads above organic results. Bid-based.

SEO requires both marketing AND engineering effort (e.g. Skyscanner needed hundreds of thousands of landing pages). This is another example of Conway's Law.

Awareness-Based Channels

Users do NOT have current intent. The goal is brand awareness so users remember you when they do have intent. **Much lower conversion rates** than intent-based channels. Examples: TV ads, Facebook/Instagram ads.

Facebook's targeting capabilities are extremely granular (location, age, interests, behaviour, etc.), making it more effective than traditional awareness channels. But awareness-based marketing is expensive and many startups skip it entirely.

Why use awareness channels despite low conversion? They build brand awareness and improve the performance of intent-based channels (users recognise and trust the brand when they search). Many companies blend both types.

Referral-Based Channels

Users arrive via CGMs/invitations from other users. Key differences: (1) can drive compounding, exponential growth, (2) most have no marginal cost per user acquired (unlike paid channels).

EXAM TIP: A common question: 'Explain why intent-based channels have higher conversion than awareness-based.' Answer: because intent-based channels reach users who are already searching for a solution, while awareness-based channels reach users without current intent. But awareness channels are still used to build brand awareness and improve performance of intent-based channels.

5.2 The Four Fits Framework

Four 'fits' must be considered when setting acquisition strategy:

1. Product-Market Fit

Does the market want and adopt your product?

2. Business-Model <-> Market Fit

Can the business sustainably meet its revenue target from this market? Revenue target = (number of addressable customers) x (revenue per customer). Example: WhatsApp could realistically find 100M users at \$1/year, but NOT at \$100/year. Many startups get this wrong by overestimating market size or revenue per customer.

3. Business-Model <-> Channel Fit

Can the business model afford the channel? E.g. if you charge \$1/year per user, you cannot afford an expensive sales team (which is why WhatsApp used referral-based acquisition).

4. Channel <-> Product Fit

Is the channel appropriate for the product and its market? E.g. a product aimed at teenagers will not succeed using email marketing.

5.3 Acquisition Metrics: CPA, CAC, LTV

CPA — Cost per Acquisition

Average cost to acquire a new user through a specific channel. Includes ONLY channel costs (ad spend, etc.). Does NOT include activation/conversion costs. Good for comparing raw channel performance.

Warning: always segment by channel. A blended CPA can hide one great channel and one terrible one.

CAC — Cost of Customer Acquisition

$$\text{CAC} = (\text{Marketing costs} + \text{Additional conversion costs}) / \text{Number of customers acquired}$$

Includes ALL costs from acquisition through to first revenue: channel spend (CPA), sales team salaries, engineering costs for activation funnels, etc.

Common mistakes: (1) excluding costs to make CAC look lower (self-deception), (2) using blended CAC that hides poor-performing channels, (3) not segmenting by customer type.

LTV — Customer Lifetime Value

$$\text{LTV} = \text{Average revenue per customer per period} \times \text{Retention period}$$

How much revenue a customer generates over their entire lifetime of using the product. Better retention = higher LTV.

LTV:CAC Ratio

- **>= 3:1** — Healthy. Acquisition is profitable with headroom for fixed costs.
- **1:1 to 3:1** — Profitable on acquisition but may not cover overheads.
- **< 1:1** — Losing money per customer. Only acceptable temporarily (to seed CGMs or run experiments).

Best practice: Calculate LTV assuming only 6-month retention until data proves otherwise. This prevents dangerous over-optimism. Many startups assume 4-year LTV with only 1 year of data — this is self-deception.

EXAM TIP: LTV:CAC calculations are very likely on the exam (15 marks on acquisition). ALWAYS show your working. Include ALL costs between acquisition and revenue. Convert annual salary costs to monthly per-customer figures. State your conclusion about whether the ratio is healthy.

5.4 Time Displacement

Sometimes there is a lag between marketing spend and revenue generation. If the lag exceeds one measurement period, use: $CAC = (\text{Sales and Marketing expenses from } N \text{ months ago}) / (\text{customers converted this month})$, where N is the lag period.

5.5 Super-Aggregation

An **aggregator** is the 'point-of-discovery' for a category (e.g. Booking.com for hotels). A **super-aggregator** has become so dominant that businesses change their own practices to fit the aggregator's requirements (e.g. Google). This drives up advertising costs on these platforms, making it harder for startups to compete with paid acquisition — another reason to invest in CGMs.

Chapter 6: Compounding Growth Mechanisms (CGMs)

6.1 What is Compounding Growth?

A product experiences compounding growth when the **rate of growth at any point is proportional to the number of users at that point**. If sustained, this leads to accelerating (exponential) growth.

Compounding growth is driven by two separate forces: **CGMs** (referral-based) and **Network Effects** (attraction-based). These are NOT the same thing, but they can reinforce each other.

6.2 What is a CGM?

A CGM is a mechanism where the output drives the input in a repeating loop. In product terms: existing users → mechanism → new users → those new users feed back into the mechanism. The more users in, the more additional users out.

The **Referral** state in the customer journey is the link between the customer journey and CGMs. Users can only refer others if they are retained and using the product.

6.3 CGM Types — Complete Taxonomy

Direct CGMs (currency = users)

Users directly invite other users.

- **Personal Utility:** The referrer gains product value from the referral. E.g. inviting friends to WhatsApp so you can message them. MOST desirable because motivation is strongest and marginal cost is near zero. But not all products can support this.
- **Financial:** Monetary incentive for referral. E.g. 'Refer a friend, both get £10.' Works best when both referrer and referred user benefit. Less attractive because each new user costs money. Important: the financial incentive must be sustainable relative to customer LTV (Business-Model ↔ Channel Fit).
- **Social Capital:** Users gain status/reputation from referring. E.g. being seen as a helpful recommender. Often acts alongside other CGM types. Primary motivation question: 'What is the main driver for the referral?'

KEY CONCEPT: Financial Asymmetry: In financially-incentivised CGMs, the perceived value of the incentive to the user can be greater than the actual cost to the business. E.g. offering a free month of service costs the business only the marginal cost of serving that user, not the subscription price.

Indirect CGMs (currency ≠ users)

Content-Based (currency = content):

- **Company-Generated Content:** Easier to start (company decides to write), harder to scale (limited by staff). Can control quality and relevance.
- **User-Generated Content:** Harder to start (need to persuade users), easier to scale (scales with user base). Creates a truly compounding mechanism.

- **Combined strategy:** Start with company-generated content, transition to user-generated as the user base grows. This combines the advantages of both.

Indirect CGMs can also act as **retention/awareness mechanisms** because they remind users the product exists between buying events.

Money-Based (currency = money):

- **Paid-Ads CGM:** Revenue from users → buy ads → ads bring new users → some sign up → generate revenue → buy more ads.
- **Sales-Team CGM:** Revenue from users → pay sales team → sales team closes new users → generate revenue → hire more salespeople.
- **Combined:** Use paid ads or cheap prospecting reps to generate leads, then hand leads to more expensive sales executives to close. Common in B2B.

IMPORTANT: Money-based CGMs must be carefully checked for affordability. Often the LTV of customers does not cover these acquisition costs.

6.4 CGM Lifecycle — Three Phases

Phase 1: Inertial Phase

At the outset, it is difficult to get the CGM moving. Users are reluctant to refer an unknown brand (reputation risk). Invitees are reluctant to join a product with few members (low perceived value). In 2-sided networks, insufficient suppliers discourage customers and vice versa.

Tactics to overcome inertia:

- **Geo-concentration:** Focus efforts on one city/area to build critical mass.
- **Subsidise one side:** E.g. Uber subsidised drivers in new cities.
- **Unsustainable marketing spend:** Temporarily over-invest to create a critical mass of users.
- **Build atomic networks:** Start with small, dense networks (e.g. Facebook → one university at a time).

Phase 2: Growth Phase

The CGM is operating well and the compounding effect is visible. This is where the quantitative CGM model applies (see Chapter 8).

Phase 3: Starvation Phase

The CGM runs out of 'fuel' — potential users that can be reached by the mechanism are saturating. Later referrals go to weaker connections with lower conversion. Performance drops.

Action: Ramp down optimisation of this CGM. Redirect resources to establishing new CGMs, moving into adjacent markets, or other significant changes.

EXAM TIP: The CGM lifecycle is a very common exam topic. Be ready to: (1) explain the inertial phase and give two examples of tactics to overcome inertia (from different businesses), (2) explain starvation and what action to take, (3) calculate the Growth Multiplier in the growth phase.

6.5 Network Effects vs CGMs

This is a critical distinction — they are NOT the same mechanism:

- **Network Effects** attract participants through the VALUE of the network. Not referral-based. Will operate even if no referral mechanisms exist.

- **CGMs** are referral-based mechanisms. Users refer other users through direct or indirect actions.
- CGMs can be **accelerated** by network effects (more users → more referrals).
- CGMs can **accelerate** network effects (more referrals → more users → stronger network).
- But they can **operate independently** of each other.

Chapter 7: Network Effects

7.1 Definition

When the greater the number of participants in a network, the greater the value of the network to those participants, this acts as an attractive force to bring new participants. This is described by **Metcalfe's Law: V is proportional to n -squared**, where V = network value and n = number of nodes.

Implication: network effects are **insignificant when the network is small** but become very powerful at scale. This is why products relying only on network effects (without CGMs) struggle to scale from a low user base.

7.2 Types of Network Effects

1-Sided Networks

All participants are the same type. E.g. Facebook before advertisers — more friends on Facebook = more valuable for you.

2-Sided Networks

Two participant types that attract each other. Common configuration: buyers and sellers. Buyers attract sellers, sellers attract buyers. Examples: Uber (riders/drivers), Airbnb (guests/hosts), eBay (buyers/sellers), Booking.com (travellers/hotels).

N-Sided Networks

Often evolve from 2-sided networks. Additional participant type is usually advertisers or content providers. Example: Facebook (users + advertisers + content creators).

Scale-Based Network Effects

As a network scales, additional effects emerge:

- **Data network effects:** More usage data → better product optimisation → more value.
- **Brand effects:** 'If everyone uses it, it must be trustworthy.'
- **Purchasing power:** Buy in bulk → secure discounts → share (or retain) savings.

7.3 Network Liquidity

In a 2-sided network, if suppliers find too few buyers (or vice versa), they leave. Early-stage networks must subsidise one side to attract the other. E.g. Uber subsidises drivers in new cities. At scale, manage liquidity with mechanisms like surge pricing.

7.4 Bootstrapping Strategies

- **Focus on the 'hard side':** One side of the network produces most value. Identify and nurture it. E.g. suppliers in a marketplace, active content creators in social networks.
- **Build atomic networks:** A small, dense network is more valuable and stronger than a large, sparse one. Start with tight-knit subsets and grow outward. E.g. Facebook started with one university at a time.

KEY CONCEPT: Atomic vs Sparse Networks: Google+ moved billions of Gmail users over — massive numbers but no meaningful connections. It collapsed because the network was sparse. Facebook succeeded by building dense atomic networks (universities) first. Threads (Meta) had a similar issue — 100M users in 30 days from Instagram, but Instagram connections did not equal Twitter-like connections.

7.5 Anti-Network Effects

As networks grow, they attract forces that compete against network effects and can slow or reverse growth:

- **Bad netiquette:** New users don't know the social rules.
- **Spam and trolling**
- **Fake news**
- **Hard-side rebellion:** Most valuable participants protest (e.g. Uber's full-time drivers protesting conditions).
- **Enshittification:** Self-inflicted harm by chasing revenue at the expense of user experience.

Growing a network is a battle between network effects and anti-network effects.

7.6 Trust in Networks

Networks require trust mechanisms. E.g. Airbnb's mutual rating system. Amazon's buyer protection. Without trust, network participants are reluctant to transact, which reduces liquidity and value.

7.7 Supply-Side CGMs

CGMs **rarely work** on the supply side of 2-sided networks because suppliers compete with each other — no incentive to refer competitors. Exception: works when suppliers believe demand is large enough that new suppliers won't hurt them (e.g. Uber drivers incentivised to recruit other drivers when demand is high).

Chapter 8: Quantitative Growth Models

8.1 Overview

Quantitative growth models come in two types: **component models** (focus on one part of the growth engine) and **integrated models** (bring everything together). Component models cover: CGMs, Retention, Activation, and Acquisition.

8.2 CGM Quantitative Model — Growth Multiplier

To model a CGM quantitatively, tabulate each sub-step of the referral loop. Each step has a conversion parameter (what percentage of users proceed to the next step). Apply these parameters iteratively:

Worked Example: A product has a direct CGM where users scan contacts, send invitations, and recipients may click, visit, and sign up.

Parameter	Value
Scan Contacts	20%
Send Invites	60%
Invitations Sent	35
Click	70%
Visit	40%
Sign-up	40%

Starting with 1000 users in Cycle 1:

- Cycle 1: 1000 sign-ups → 200 scan contacts → 120 send invites → 4200 invitations → 2940 clicks → 1176 visits → **470 new sign-ups**
- Cycle 2: 470 → ... → **221 new sign-ups**
- Continue until the CGM exhausts referral capacity...
- Total sign-ups across all cycles: **1887**

So the CGM turned 1000 users into 1887 total users — **887 'free' users**.

The Growth Multiplier Formula

$$V = (\text{Sign-ups in Cycle 2}) / (\text{Sign-ups in Cycle 1})$$

$$\text{Growth Multiplier (GM)} = 1 / (1 - V)$$

In our example: $V = 470/1000 = 0.47$, so $GM = 1/(1-0.47) = \mathbf{1.89}$

The CGM delivers 1.89x the starting number of users. This multiplier applies to ALL users fed into the mechanism.

Why Small Improvements Compound

Changing just one parameter by 2 percentage points (Scan Contacts from 20% to 22%) takes GM from **1.89 to 2.07** — a significant uplift from a tiny change. This is the power of compounding: small improvements in any sub-step amplify through the entire loop.

Attenuating Factor

In the inertial and starvation phases, the CGM performs below its growth-phase level. An attenuating factor (0 to 1) can be applied to the model to reflect this. During growth phase, the attenuator is 1.0. During inertia, it might be 0.1-0.4. During starvation, it decreases from 1.0 back toward 0.

8.3 Experimentation and Prioritisation

Constraint Analysis

'Which is the thinnest pipe?' In any flow-based system, one constraint determines the throughput. Find that constraint and optimise it first. There is only one constraint at any point in time (unless two are identical).

ICE Framework

Score proposed experiments on three dimensions (each 1-5):

- **Impact:** How great is the expected positive effect? (1=low, 5=high)
- **Confidence:** How certain are we the impact will be realised? (1=low, 5=high)
- **Effort:** How much effort to implement? (1=very high effort, 5=very low — REVERSED)

High total ICE score → prioritise first. Alternatively, plot Impact vs Risk on a 2x2 chart. Focus on high-impact, low-risk experiments first.

Optimisation Cycle

Generate Hypothesis → Prioritise (ICE/Constraint) → Test → Learn → Repeat. This cycle runs continuously across all parts of the growth model.

EXAM TIP: Growth Multiplier calculations may appear on the exam. Know how to: (1) calculate V from the ratio of two successive cycles, (2) apply $GM = 1/(1-V)$, (3) explain why small improvements have outsized effects due to compounding. Show ALL working.

Chapter 9: Integrated Growth Models

9.1 Purpose

Component models analyse one part in detail. Integrated models bring everything together to: (1) understand how components interact, (2) perform forward predictions, (3) run what-if scenarios, and (4) build credibility with investors.

Integrated models are typically built in spreadsheets. Each component gets a simpler version of its detailed component model.

9.2 Building Steps

- **Step 1:** Build the Integrated Qualitative Model first — map all network effects, CGMs, retention mechanisms, and linear acquisition channels.
- **Step 2:** Define model outputs in a summary 'Overall View' tab.
- **Step 3:** Build Acquisition tab — linear acquisition sources + CGM-based acquisition.
- **Step 4:** Build Retention tab — starting active users + new acquisitions - churn. Include activation rate.
- **Step 5:** Build Monetisation tab — revenue from different user tiers.
- **Step 6:** Build Costs tab — marginal costs per user.
- **Step 7:** Connect everything to Overall View.

9.3 Key Features

The model is NOT a P&L; statement. Its currency is mainly users, not money. Where costs are modelled, these are marginal costs, not fixed costs. But a growth model can feed into a P&L; for more accurate budgeting.

9.4 Moving Variables

Variables in the model can change over time. Five possibilities: improve over time, diminish over time, improve then diminish, diminish then improve, or stay the same. Adding moving variables makes predictions more realistic.

Example: as a brand becomes more trusted, organic acquisition may become easier (improve). But as the market saturates, acquisition becomes harder (diminish). Combining these: improve then diminish.

EXAM TIP: For the exam, know the components of an integrated model (acquisition, activation, retention, monetisation, costs), understand that CGMs are incorporated as an acquisition source, and know what moving variables are and why they matter.

Chapter 10: Organisation and Scaling

10.1 Organisational Inflexion Points

As a company grows, outcomes rise in line with headcount until approximately **30-50 people**, when a sudden inflexion point occurs. Outcomes stall despite continued hiring. This pattern repeats at ~100, ~200, ~400+ people.

The cause: people, processes, and structures that worked at one scale become inappropriate at a larger scale. All three often scale out simultaneously, intensifying the impact.

10.2 Agency — The Core Optimisation Principle

When designing or changing organisations, the answer to 'what are we optimising for?' is **agency** — the ability of each employee to fulfil the responsibilities of their role.

In cities, productivity per person increases with size (natural tendency). In companies, it often decreases. The difference is that companies inadvertently destroy agency as they grow.

The Four Components of Agency

- **1. Competence:** Is the employee competent in their role? Competence degrades with scale unless invested in. E.g. a great engineer promoted to manage 5 people may be immediately incompetent at leadership. Plan competence needs at least a year ahead.
- **2. Ownership:** Is the employee clear about what decisions they own? As companies grow, leaders hold on to ownership too long. Decisions concentrate at the centre. Decision latency increases. Decision intensity drops proportional to the SQUARE of decision latency.
- **3. Alignment:** Is the employee aligned to company goals? Too little alignment → wrong decisions. Too much alignment → excessive process overhead squeezes out time for actual work. Find the optimum.
- **4. Access to Resources:** Can the employee access the skills and resources needed? Functional silos often prevent access across teams. Organisation design must ensure cross-functional resource access.

KEY CONCEPT: Decision Intensity: The speed of the business depends on how many decisions are made at the front line. If decisions are centralised, decision intensity drops proportional to the SQUARE of decision latency. Push decisions to the front line.

10.3 The Spotify (Squads and Tribes) Model

Small teams (squads) of ~8 people that emulate startups within the larger company: multi-disciplinary, autonomous, with their own mission, metrics, and roadmap. Squads are organised into Tribes for coordination. This model attempts to recreate the high agency of small startups at larger scale.

EXAM TIP: A likely exam question: CEO complains engineers are slower than when the founders did everything. Her conclusion: poor hiring. Your response: more likely a SYSTEM-level problem. Check agency: (1) ownership clarity, (2) competence investment, (3) alignment, (4) resource access. Then check processes. Only then consider individual capability.

10.4 McNamara Fallacy

Making decisions based solely on quantitative metrics while ignoring qualitative factors. Named after Robert McNamara's approach during the Vietnam War (measuring success by enemy body counts while ignoring political and social realities). In startup context: over-reliance on easily measurable metrics while ignoring important things that are harder to measure. Related to 'making the measurable important rather than making what's important measurable.'

Chapter 11: Investment Fundamentals

11.1 Why Tech Startups Need Investment

Tech businesses have low marginal costs at scale (cost of onboarding a new Google user is ~zero). But reaching that point requires high upfront R&D; costs. Network effects also encourage rapid scaling beyond the natural rate, requiring capital.

11.2 Investment Stages

Stage	Typical Size	Company Stage
Pre-Seed	<\$100K	Idea/very early
Seed	<\$2M	Seeking PMF
Series A	~\$10M	Driving growth
Series B	\$30M+	Further growth at scale
Series C+	Larger	Rapid scaling / pre-IPO

Angels invest at seed stage (individuals, often former founders). **VCs** invest at later stages using equity stakes.

11.3 How VC Investment Works

- VC raises a **Fund** from external investors (**Limited Partners** — pension funds, endowments).
- VC = **General Partner**, manages the fund for 5-10 years.
- Invests in a portfolio of startups. Most will fail. A few must return **10x+** to cover all losses and deliver profit.
- Upon exit (acquisition or IPO), VC returns original investment to LPs, takes **Carried Interest** (performance fee from profits), then returns remaining profit to LPs.

11.4 Dilution

Each funding round issues new shares, diluting existing shareholders' percentage ownership. Example: founder owns 10% (100k of 1M shares). Investor buys 10% for £100k, receiving 100k new shares. Total shares: 1.1M. Founder now owns $100k/1.1M \approx 9\%$. Founders must ensure company valuation keeps pace with dilution.

Preference Shares: Investors sometimes secure terms giving them disproportionate share of exit proceeds. Preference shareholders are paid first, which may leave little for ordinary shareholders (founders, employees).

Chapter 12: AI Impact on Growth Engineering

This topic is exam-relevant because ChatGPT is a confirmed exam product.

12.1 Impact on Content-Based CGMs

- **Positive:** Companies can generate content in greater volumes more quickly using AI, extending company-generated content CGMs before needing user-generated content.
- **Negative:** AI-generated content may be lower quality, repetitive, and users may object to reading it. 'AI slop.'
- **Hybrid approach:** Auto-generate content that prompts users to comment. Combines AI scalability with human freshness.

12.2 The Search-to-Answer Engine Shift

LLMs are fundamentally changing how people consume content:

- Google: 2 pages scraped per 1 visitor (pre-AI) → 6:1 (late 2024) → **18:1 (mid 2025)**. Cause: AI Overviews.
- ChatGPT: 250:1 (late 2024) → **1500:1 (mid 2025)**. People trust AI summaries and don't click through.

Implications: content-based CGMs becoming less effective. SEO declining → more businesses buy ads → higher ad costs → money-based CGMs harder to operate profitably.

Content-based CGMs still have value as **retention mechanisms within a platform** (e.g. Medium, LinkedIn, Instagram content retains users on the platform).

Chapter 13: Exam Product Guides

These four products are confirmed to appear in the exam. Think through the growth model for each.

13.1 ChatGPT

- **Network effects:** Data network effect (more users → better model), 2-sided (users + developers/plugin creators).
- **CGMs:** Content-based indirect (users share outputs, screenshots go viral), organic word-of-mouth.
- **Exam angle:** Impact on content-based CGMs, search→answer engine shift, AI overviews reducing traffic.

13.2 Snapchat

- **Network effects:** 1-sided (more friends = more value).
- **CGMs:** Direct personal-utility (invite friends to chat), content-based indirect (Stories shared externally).
- **Retention:** Streaks feature as engagement/retention mechanic. Social media products should retain ~60% after activation.
- **Activation:** Needs friends on platform to deliver value — cold start problem.

13.3 Medium

- **Network effects:** 2-sided (writers + readers).
- **CGMs:** Content-based indirect (articles shared via social/SEO attract readers). User-generated content model.
- **Exam angle:** CG vs UG content, content CGM lifecycle, AI impact on content platforms.

13.4 Food Delivery (Deliveroo / Uber Eats / Meituan)

- **Network effects:** 2-sided or 3-sided (diners + restaurants + riders). Strong liquidity requirements.
- **Inertial phase:** Geo-concentration (city by city), subsidise restaurants/riders, build atomic networks in neighbourhoods.
- **Retention metric:** e.g. 4o28 (4 orders every 28 days).
- **CGMs:** Financial referral ('refer a friend, both get £10'), organic word-of-mouth.
- **Swipii-type question:** Focus on one city — referral works better with critical mass, easier to create atomic network, easier to test and learn.

Chapter 14: Practice Questions

Work through these questions without looking at the answers. They are arranged by topic and increase in difficulty. Check your answers in Chapter 15.

Retention and Activation

[Easy] Q1 (3 marks)

What are the three typical patterns shown on a retention-cohort graph? For each, state what it indicates about the business.

[Easy] Q2 (4 marks)

Explain the structure of a retention metric (NaP). Give two rules for setting a good NaP metric and explain why each rule matters.

[Easy] Q3 (2 marks)

A tax return software company uses the retention metric 4r28 (4 tax reviews every 28 days). Comment on the suitability of this metric.

[Medium] Q4 (2 marks)

Explain the Activation Inequality. How do growth teams use it to optimise onboarding?

[Medium] Q5 (2 marks)

A startup's activation workflow requires users to grant access to their phone contact list and invite friends BEFORE they can fully use the app. Discuss the pros and cons.

[Hard] Q6 (4 marks)

Seven sequential cohorts are shown. Cohorts 1-3 improve steadily. Cohort 4 performs badly during activation but then partially recovers. Cohorts 5-7 resume the improvement trend. Comment on the sequence and explain the Cohort 4 anomaly.

[Medium] Q7 (3 marks)

A delivery app uses 4o28 as its retention metric. Describe how you would categorise users into Power, Core, and Churn-risk categories. What action should the team take when a user moves from Core to Churn-risk?

Compounding Growth Mechanisms

[Easy] Q8 (3 marks)

Explain the difference between Direct and Indirect CGMs. Give one example of each type from a real product.

[Medium] Q9 (3 marks)

Explain what is meant by the Inertial Phase of a CGM. Provide one example from each of two separate businesses of approaches to overcome this inertia.

[Easy] Q10 (2 marks)

Why do growth teams sometimes use company-generated and user-generated content-based CGMs in sequence?

[Medium] Q11 (2 marks)

Explain the relationship between network effects and CGMs.

[Medium] Q12 (3 marks)

A CGM has $V = 0.52$. Calculate the Growth Multiplier. If an experiment improves V to 0.58, what is the new GM? Comment on the significance of this improvement.

[Hard] Q13 (2 marks)

A growth team constructs a quantitative model for an n-sided network-based trading platform. Explain what future trend you would incorporate for linear acquisition.

Acquisition

[Easy] Q14 (3 marks)

Describe the three main types of acquisition channels.

[Medium] Q15 (2 marks)

Explain why intent-based channels usually exhibit a higher conversion rate than awareness-based channels. Why are awareness-based channels still used?

[Hard] Q16 (4 marks)

A startup acquires 1500 customers/month at \$80 CPA. Sales team costs \$120,000/year. 6-month LTV is \$300. Comment on the acquisition strategy.

[Medium] Q17 (4 marks)

You join a startup as Growth Director. The team boasts about an LTV:CAC of 6:1. What four questions would you ask to determine the accuracy of this claim?

[Easy] Q18 (1 marks)

In what circumstances is it appropriate to operate with $CAC > LTV$?

Network Effects

[Medium] Q19 (3 marks)

Explain how a startup should manage liquidity in the earliest stages of a 2-sided network.

[Medium] Q20 (3 marks)

Explain the operation of an n-sided network with an example.

[Medium] Q21 (2 marks)

Swipii (a cashback app) is a two-sided network of shoppers and shops. When entering a new country, should they target shops across many cities or focus on one or two cities? Explain.

[Easy] Q22 (1 marks)

Why do CGMs rarely work on the supply side of 2-sided networks?

Organisation

[Medium] Q23 (3 marks)

A founding team of a small startup iterates rapidly by making all decisions. When the team grows to 40 people, iteration speed drops considerably despite working the same way. Explain the most probable cause.

[Hard] Q24 (3 marks)

A founder-CEO laments that engineers are slower than when she and co-founders did all the work. She concludes the problem is hiring quality. Comment on her conclusion and suggest what she should check first.

Growth Model Application

[Hard] Q25 (5 marks)

Using your knowledge of Disney+, describe its growth model. Include at least four distinct mechanisms.

[Medium] Q26 (2 marks)

Explain why a product with very strong organic word-of-mouth CGMs might not need classical linear marketing, while most startups do.

Chapter 15: Answers to Practice Questions

A1:

(1) Green line — drops then levels off. Indicates a viable business; users have formed habits and retention has stabilised. If most cohorts look like this, the business has PMF. (2) Yellow line — continues falling slowly. Indicates 'work to do'; retention is not catastrophic but needs improvement. Common in early-stage startups. (3) Red line — falls rapidly toward zero. Indicates the business is heading for failure unless retention is significantly improved.

A2:

The retention metric has the form NaP: N = number of times an action is performed, a = the specific action, P = the time period in days. Two rules: (1) N should be greater than 1, because this gives a better signal-to-noise ratio and reduces false churn signals (if N=1, a single missed action triggers a false alarm). (2) P should be greater than the typical usage cycle of the product. For example, if the product is used mainly on weekdays, setting P=1 day would create false churn signals on weekends, obscuring real retention trends.

A3:

This is an inappropriate metric for most months of the year because tax products are typically used yearly (around tax season). Both the number (4) and the period (28 days) are far too frequent for the product's natural usage frequency. However, during the tax return season itself, this metric might be more appropriate as users may review their tax details multiple times over a few weeks.

A4:

The Activation Inequality states: Perceived Value must be greater than Perceived Effort. If the effort required to start using the product (signing up, entering details, learning the interface) exceeds the value the user perceives, they will be lost before activating. Growth teams use this concept to optimise the ordering of steps in the activation process and to minimise the amount of information requested from users before delivering value to them. Value demonstration should come before heavy information gathering.

A5:

Pros: (1) Likely to drive higher referral levels because every user must go through the referral step. (2) Even users who don't fully activate will still have contributed to the referral process, generating value from users who would otherwise be lost. Cons: (1) Increases friction in the activation process, likely reducing the activation rate (violates Activation Inequality by adding effort before value). (2) Delays providing value to the user, which also damages activation. The key trade-off is between referral volume and activation rate.

A6:

In general, cohorts are improving over time, likely due to ongoing product improvement through experimentation (1 mark). However, Cohort 4 is out of sequence — it performs relatively badly during activation and then partially recovers (1 mark). Possible explanations: a bug or problematic change was temporarily introduced into the activation flow, or an external event occurred (holiday, service outage, seasonal effect) that distracted users or

made them less likely to activate. This is supported by the fact that later cohorts resume the improvement trend, and some Cohort 4 users who didn't activate initially appear to move out of dormancy later (2 marks).

A7:

Power user: (6+)o28 — 6 or more orders every 28 days. These are the most valuable users with the deepest habits. Core user: (4-5)o28 — meeting or slightly exceeding the base retention metric. These users are healthy and likely to be retained. Churn-risk: (<4)o28 — below the retention metric threshold. When a user drops from Core to Churn-risk, this should trigger immediate intervention: targeted offers, push notifications, personalised recommendations, or discounts to re-engage them before they churn entirely.

A8:

Direct CGMs: users directly invite other users. The 'currency' is users themselves. Example: WhatsApp — users invite friends to the app so they can message them (personal utility-based). Indirect CGMs: some intermediate currency (content or money) is used to attract new users; the user may not even be aware they're attracting new users. Example: Medium — users write articles that are shared via social media and discovered via SEO, attracting new readers to the platform (content-based indirect CGM).

A9:

The Inertial Phase is the beginning of a CGM's lifecycle where it is difficult to establish user-to-user referral. Users are reluctant to refer an unknown brand (reputation risk), and invitees are reluctant to join a product with few members (low perceived value). Example 1: Uber employed geo-concentration, supply-side subsidies, and unsustainable marketing spend in new cities to create a critical mass of drivers and riders sufficient to overcome inertia. Example 2: Electric scooter startups (e.g. Lime) concentrated their entire fleet in one small area of a city, creating visible density and critical mass before expanding to other areas.

A10:

There are two sub-types of content-based indirect CGMs: Company-Generated (CG) and User-Generated (UG). CG content is easier to get started (the company can decide to produce content immediately) but harder to scale (limited by staff capacity). UG content has the opposite characteristics — harder to start (need to persuade users to create content when few others are reading) but easier to scale (grows with the user base). Using them in sequence — starting with CG to build initial audience, then adding UG as the user base grows — combines the advantages of both approaches.

A11:

Network effects and CGMs are NOT the same mechanism. (1) Network effects attract participants through the increasing value of the network as it grows. This attraction is NOT a referral mechanism. (2) CGMs are referral-based mechanisms. They can be accelerated by network effects (more users in the network → more potential referrers), they can accelerate network effects (more referrals → more users → stronger network), but they can operate independently of network effects. A product with no network effects can still have effective CGMs, and vice versa.

A12:

With $V = 0.52$: $GM = 1/(1-0.52) = 1/0.48 = 2.08$. With $V = 0.58$: $GM = 1/(1-0.58) = 1/0.42 = 2.38$. The improvement from 2.08 to 2.38 (a 14% uplift in the growth multiplier) resulted from only a 6-percentage-point increase in V . This demonstrates the power of compounding — small improvements in CGM conversion parameters amplify through the loop to produce disproportionately large improvements in overall growth performance.

A13:

The linear acquisition variable should be modelled as a moving variable with two phases: (1) As network effects strengthen with more participants, linear acquisition performance should improve — it becomes easier to attract new users because the platform is more valuable and well-known. (2) Eventually, as the market begins to saturate, acquisition performance would stabilise and then decline as fewer potential users remain. The moving variable should reflect this 'improve then diminish' pattern.

A14:

(1) Intent-based channels: users are actively searching with intent to buy or complete an action (e.g. Google Search, SEM). (2) Awareness-based channels: users do not have current intent; the goal is brand awareness for future recall (e.g. Facebook ads, TV advertising). (3) Referral-based channels: users arrive via invitations or referrals from existing users through CGMs.

A15:

Intent-based channels have higher conversion because users are already searching for a solution when they discover the product. In awareness-based channels, users see ads without current intent, so conversion is naturally lower. However, awareness-based channels are still used because they build brand awareness and reduce the likelihood that users forget about the product. This often improves the performance of intent-based channels too, because users recognise and trust the brand when they come to search.

A16:

Calculate LTV:CAC. 6-month LTV = \$300. Channel CPA = \$80. Sales team cost per customer = $\$120,000 / (12 \text{ months} \times 1,500 \text{ customers/month}) = \6.67 . Total CAC = $\$80 + \$6.67 = \$86.67$. LTV:CAC = $\$300 / \$86.67 = 3.46$. The ratio exceeds 3:1, which means the acquisition is profitable. The headroom above 3:1 means the acquisition process generates profits likely to cover other fixed costs in the business. This acquisition strategy is sound.

A17:

Four questions: (1) What costs are included between acquisition and revenue generation? Are all conversion costs (sales team, engineering, support) included, or just channel spend? (2) Is the CAC segmented by channel? A blended 6:1 might hide one channel at 10:1 and another at 0.5:1. (3) Is the CAC segmented by customer type? Different customer segments may have very different CACs. (4) What assumptions underlie the LTV? What revenue per customer and what retention period are being assumed? What data supports those assumptions? (Many startups assume unrealistically long retention periods.)

A18:

It is appropriate to operate with $CAC > LTV$ for limited periods in order to: (a) generate sufficient users to conduct statistically valid experiments, or (b) 'spike' a CGM in its inertial phase to build critical mass. Both are strategic

investments that should be time-limited with clear objectives.

A19:

Liquidity in a 2-sided network means ensuring sufficient and balanced supply and demand. In early stages, there is usually too little of both. For the supply side: incentivise suppliers to join (e.g. guaranteed minimum earnings), compensate them for lack of demand, actively recruit high-quality suppliers. For the demand side: drive referral mechanisms with incentives, invest in advertising (potentially at unsustainable levels), offer discounts or free trials to early users. The goal is to reach a point where organic network effects take over from subsidised growth.

A20:

In an n-sided network, there are n types of participant. One type of participant acts as an attractor for other types, and vice versa, creating self-reinforcing network effects. Example: Facebook has three sides — users, advertisers, and content creators. More users attract more advertisers (who want to reach that audience) and more content creators (who want readers). More content attracts more users. More users and content attract more advertisers. These three sides reinforce each other, strengthening the network.

A21:

Swipii should focus on one or two cities at a time, for two main reasons: (1) Referral mechanisms work better when users reach critical mass, which is more achievable when all efforts are concentrated in one city. (2) As a two-sided network, it is easier to create an atomic network in one city — building density of both shoppers and shops in a small area first, proving the product works, and then expanding. Spreading thinly across many cities would create sparse networks on both sides with insufficient liquidity.

A22:

On the supply side of 2-sided networks, participants are often in direct competition with each other. There is little or no incentive for a supplier to refer other suppliers, as doing so would attract competition. Exception: supply-side CGMs can work when suppliers believe demand is large enough to accommodate new suppliers without negative impact on the referring supplier.

A23:

The business is slowing down because decisions are being overly centralised at a scale not appropriate to this model. The founding team made all decisions effectively at 5 people, but at 40 people, the same approach creates long decision queues. Decision latency increases, which reduces decision intensity (proportional to the SQUARE of decision latency). This affects employee morale and sense of agency, further reducing iteration speed. Eventually the founders become overloaded, worsening the problem. The solution: push agency to the front line of the business as a strategic goal.

A24:

The CEO may be correct, but it is more likely a system-level problem caused by organisational growth. She should first assess agency: (1) Is everyone clear about their job and what decisions they own? (2) Are people competent in their current roles — remembering that competence degrades with scale unless invested in? (3) Are people well-aligned to the strategy and goals? (4) Do people have access to the skills and resources they need?

She should then check whether key processes have degraded. Only if all of the above checks out well should she consider individual staff capability — some people may have 'scaled-out' and struggle to operate effectively in a larger organisation.

A25:

Disney+ growth model mechanisms: (1) Aggregator Effect: Disney+ is the only streaming service where all Disney content can be viewed, attracting viewers who like Disney content. More content created strengthens this effect. (2) Organic Direct CGM: Users pay subscription → Disney makes new content → users watch and tell friends via word-of-mouth → some friends sign up. (3) Content-based Indirect CGM: Users can share accounts with up to 7 people (4 simultaneous devices). This creates eventual pressure for some users to sign up for their own account. (4) Money-based Indirect CGM: Subscription revenue funds advertising spend to acquire more subscribers, seeding other CGMs. (5) Activation/Engagement: Free trial periods demonstrate value before requiring payment. Regular new content releases, group watch features, recommendation engine, auto-play next episode, multiple user profiles for relevant recommendations.

A26:

Linear marketing provides initial users to 'seed' CGMs — without it, CGMs have no users to operate upon. The analogy is that linear marketing is the 'flame' that ignites compounding growth, or the 'principal' upon which compounding 'interest' is earned. Products with very strong organic word-of-mouth (e.g. early WhatsApp) generate enough organic referrals that no additional seeding is needed. But most startups have CGMs that require initial user volume to achieve momentum, and organic word-of-mouth alone is insufficient. Hence most startups need linear marketing to reach the critical mass where CGMs begin to compound.

Good luck in your exam!

Remember: check marks per question, show all working, answer both sub-parts, and start with a question you can do confidently.